

Port Scanner: Step-by-Step Guide

How to run `port_scanner.py` and verify the scan results

Before you start

- Python 3 must be installed.
- The script is at `...\Network port scanner\port_scanner.py`.

Only scan hosts you own or have explicit permission to scan. Scanning others without permission can be illegal, even for "just testing." The example below uses 192.168.1.1 as an example target.

Step 1: Open a terminal in the project folder

Open the project folder in **File Explorer**. Right-click an empty area and choose **Open in Terminal** (Windows 11), or click the address bar, type **powershell**, and press Enter. Either way, the terminal opens already in the project folder, so you do not need to change directory.

Prefer to type it? Open **PowerShell** from the Start menu, then go to the folder:

```
cd "C:\path\to\Network port scanner"
```

Replace the path with the location of the project folder on your computer.

Step 2: Start the script

Click inside the terminal, type this, and press Enter:

```
python port_scanner.py
```

Step 3: Answer the three prompts

Type each value and press Enter after each one:

```
Target host/IP [127.0.0.1]: 192.168.1.1
Start port [1]: 20
End port [1024]: 80
```

The values in brackets are the default values; type a different value when needed.

Step 4: Wait for the scan to finish

The scanner displays each open port as it is found. The time required depends on the target and port range. A pause between results is normal.

Step 5: Check the result

Example output: the ports and time you see will differ. The Done : line means the scan has finished.

```
Scanning 192.168.1.1 (192.168.1.1), ports 20-80...
```

```
22/tcp open  SSH-2.0-dropbear
23/tcp open
53/tcp open
80/tcp open  HTTP/1.1 404 Not Found
```

```
Done: 4 open port(s) in 35.0s.
```

Step 6: Stop early (optional)

Press **Ctrl+C**. The scanner prints a summary for the ports it has checked so far. To scan again, repeat from Step 2.

What the example results mean

Port	Service	Notes
22	SSH	The banner in the example output shows Dropbear SSH.
23	Telnet	Unencrypted. Consider disabling it in the device's settings unless you turned it on on purpose.
53	DNS	A DNS-related service is listening on TCP port 53.
80	HTTP	An HTTP service is listening on port 80.

Good to know

- The script's defaults are target `127.0.0.1` and ports `1-1024`. Pressing Enter at every prompt uses these defaults. Scanning the full default range may take several minutes, because the script checks one port at a time and each closed port can take up to the connection timeout. Use a small range for a quick demo.
- The connection timeout is the `CONNECT_TIMEOUT` value at the top of `port_scanner.py` (0.5 seconds by default). If scans of distant or high-latency hosts miss services, consider increasing it.

Troubleshooting

Problem	Fix
python is not recognized	Try <code>py port_scanner.py</code> instead, or reinstall Python and tick "Add Python to PATH".
Could not resolve 'name'.	The host name could not be turned into an IP address (a DNS lookup failure). Check the spelling of the name and that you are connected to the network or internet. You can also enter the IP address instead.